

Every chapter so far assumes access to a scalar reward signal: dynamic programming requires $r(s, a)$, model-free RL observes r_t after each transition, and the Bellman optimality equation presupposes that rewards are known or observable. When they are neither, the DP/RL machinery cannot be applied directly.

In many domains, scalar rewards are unavailable but ordinal preferences over trajectories are easy to elicit. A human evaluator cannot assign a meaningful numerical score to a paragraph of text, but can reliably say “response A is better than response B.” The raw data is a set of trajectory pairs with binary preference labels. Reinforcement learning from human feedback (RLHF) uses these ordinal comparisons to learn a proxy reward function r_θ , which then serves as the scalar signal for standard RL optimization. This is not an inverse problem in the IRL sense; the goal is not to rationalize observed behavior, but rather a two-stage forward problem in which the analyst first learns a reward from human judgments and then solves the resulting MDP. [Christiano et al. \(2017\)](#) demonstrated that this approach could train agents without an explicit reward function. RLHF has since become the predominant method for aligning large language models.

0.1 Learning Rewards from Preferences

The canonical RLHF framework is built on a formal model of human preference. The observed data consists of tuples (s, y_w, y_l) , where s is a context, and y_w and y_l are two outputs, with y_w being the “winner” preferred by a human. Assuming preferences follow a latent utility model, the Bradley-Terry model ([Bradley and Terry, 1952](#)) gives the probability that y_w is preferred: $P(y_w \succ y_l | s) = \sigma(r_\theta(s, y_w) - r_\theta(s, y_l))$, where $r_\theta : \mathcal{S} \times \mathcal{Y} \rightarrow \mathbb{R}$ is a learned *reward model* parameterized by θ , trained to approximate human preferences (not the ground-truth reward, which is unobserved), and $\sigma(\cdot)$ is the logistic function. This formulation is a binary logit model (Section ??, Equation ??).¹ The reward model parameters θ are estimated by minimizing the negative log-likelihood of the observed human choices:

$$\mathcal{L}(\theta) = -\mathbb{E}_{(s, y_w, y_l) \sim \mathcal{D}} [\log \sigma(r_\theta(s, y_w) - r_\theta(s, y_l))], \quad (1)$$

¹[Iskhakov et al. \(2020\)](#) discuss the contrasts and synergies between machine learning and structural econometrics, including the shared reliance on logit-based choice models that underlies both RLHF and dynamic discrete choice estimation.

In the LLM setting, the “outputs” y_w and y_l are token sequences, i.e., trajectories of the autoregressive policy², so preferences are over trajectories rather than single actions. The preference loss in Equation (1) is MLE of a choice model where the “alternatives” are trajectory segments and the “choice” is the human-preferred one.

This logit foundation is also a social-choice assumption once labelers are heterogeneous. Pooling pairwise comparisons from many evaluators does not recover a neutral “human preference”; it estimates a single score whose odds ratios summarize the aggregation rule induced by the sampling scheme, label model, and loss function. Recent alignment work makes this point explicit: standard Bradley-Terry-style reward learning can behave like a hidden social welfare function, can fail natural axioms for aggregating rankings, and can create strategic reporting incentives when preferences differ across contexts or groups (Conitzer et al., 2024; Siththaranjan et al., 2024; Ge et al., 2024; Park et al., 2024). DPO inherits the same aggregation problem because it replaces the explicit reward model with an implicit reward difference, not with a theory of whose preferences count.

0.2 The RLHF Pipeline and Direct Optimization

The learned r_θ then serves as a proxy objective for policy optimization. Building on the reward-learning and RL fine-tuning framework of Ziegler et al. (2019) and Stiennon et al. (2020), the canonical three-stage pipeline was formalized by Ouyang et al. (2022). First, a base pretrained model is initialized via supervised fine-tuning (SFT)³ on a small dataset of high-quality demonstrations, yielding an initial policy π^{SFT} . This grounds the model in the desired style and format. The second step is the reward model training as described, using preference data generated from this π^{SFT} .

In the third step, the SFT policy π_ϕ is fine-tuned via PPO (Section ??) to maximize the frozen reward model r_θ ,⁴ with a KL-divergence penalty preventing the policy from

²An *autoregressive* language model generates text token by token: at each step it outputs a distribution over the vocabulary conditioned on all preceding tokens, then samples the next token; the full response is therefore a trajectory in token space and the model acts as a sequential policy over a vocabulary-sized action set.

³*Pretraining* optimizes a language model to predict the next token across a massive text corpus, producing broad linguistic knowledge with no behavioral objective. *Supervised fine-tuning* (SFT) continues training on a small curated dataset of (prompt, ideal-response) pairs to specialize the model toward the desired task and establish the reference policy π^{SFT} from which the KL penalty is measured.

⁴After fine-tuning concludes, the reward model may be reused for evaluation or for later alignment iterations, but it is not updated inside the PPO optimization loop.

drifting into regions where r_θ is unreliable. The objective is

$$J(\phi) = \mathbb{E}_{s \sim \mathcal{D}, y \sim \pi_\phi(\cdot|s)}[r_\theta(s, y)] - \lambda_{KL} \mathbb{E}_{s \sim \mathcal{D}}[D_{KL}(\pi_\phi(\cdot|s) \parallel \pi^{SFT}(\cdot|s))], \quad (2)$$

where D_{KL} is the Kullback-Leibler divergence and λ_{KL} controls the penalty strength.⁵ Without this constraint, the policy exploits inaccuracies in r_θ to achieve high proxy scores with degenerate behavior (“reward hacking”), the RLHF analogue of divergence under function approximation (Section ??).

The KL-regularized objective in Equation (2) admits a Bayesian interpretation (Korbak et al., 2022). In this view, the reference policy π^{SFT} acts as a prior distribution over plausible responses. The reward model r_θ provides evidence, specifying which responses are more desirable. The goal of alignment is to find the posterior distribution π^* that optimally combines the prior with this evidence. This ideal posterior policy takes the form of Equation (3):

$$\pi^*(y|s) \propto \pi^{SFT}(y|s) \exp\left(\frac{r_\theta(s, y)}{\lambda_{KL}}\right). \quad (3)$$

The reward function scaled by λ_{KL} defines the log-likelihood, so the KL-regularized objective $J(\phi)$ is equivalent (up to an additive constant) to the Evidence Lower Bound (ELBO) for this Bayesian inference problem. Maximizing the RLHF objective via PPO is therefore variational inference: finding the policy π_ϕ that minimizes KL divergence to π^* . This reframes the KL penalty as a structural component of the inference problem rather than an ad-hoc regularizer.

Despite this closed-form characterization, the three-stage RLHF pipeline is complex to implement, requiring training multiple large models and a computationally expensive RL loop. Direct Preference Optimization (DPO), introduced by Rafailov et al. (2023), collapses the pipeline into a single supervised learning objective by reparameterizing the reward function in terms of π^* and π^{SFT} , as in Equation (4):

$$r(s, y) = \lambda_{KL} \log\left(\frac{\pi^*(y|s)}{\pi^{SFT}(y|s)}\right) + \lambda_{KL} \log Z(s). \quad (4)$$

When this analytical expression for the reward is substituted into the Bradley-Terry preference loss from Equation (1), the unknown partition function $Z(s)$ cancels out. This

⁵ λ_{KL} denotes the KL penalty weight, reserving β for model parameters and γ for discount factors. The standard RLHF literature, including Rafailov et al. (2023), uses β for this parameter.

yields a loss function that depends only on the policy π_ϕ being optimized and the fixed reference policy π^{SFT} . The DPO objective, given in Equation (5), is thus a simple binary cross-entropy loss over policy likelihoods:

$$\mathcal{L}_{DPO}(\phi; \pi^{SFT}) = -\mathbb{E}_{(s, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\lambda_{KL} \log \frac{\pi_\phi(y_w|s)}{\pi^{SFT}(y_w|s)} - \lambda_{KL} \log \frac{\pi_\phi(y_l|s)}{\pi^{SFT}(y_l|s)} \right) \right]. \quad (5)$$

This objective is optimized on ϕ using standard supervised learning with a static preference dataset. The gradient increases the likelihood of preferred responses y_w and decreases the likelihood of dispreferred responses y_l , relative to the reference policy.

The growing family of DPO variants is best read economically by asking what part of the preference problem changes. IPO modifies the link/objective structure to avoid some DPO overfitting behavior (Azar et al., 2024). KTO replaces pairwise comparisons with a prospect-theory-style objective over desirable and undesirable outputs (Ethayarajh et al., 2024). Nash-LHF and SPPO treat preference optimization as a game over pairwise comparisons, which is closer to social choice when preferences are cyclic or non-transitive (Munos et al., 2023; Wu et al., 2024). MaxMin-RLHF makes the aggregation rule explicitly egalitarian (Chakraborty et al., 2024). ORPO, SimPO, and robust DPO mainly change implementation, reference-model dependence, or noisy-label bias rather than the underlying welfare question (Hong et al., 2024; Meng et al., 2024; Chowdhury et al., 2024). Self-rewarding loops move some feedback generation from humans to model judges, which lowers data-collection costs but makes the evaluator itself part of the mechanism (Yuan et al., 2024).

0.3 Social Choice and Whose Preferences

RLHF turns alignment into an economics problem: elicit preferences, aggregate them across people, design incentives for truthful feedback, and interpret the learned object. The phrase “alignment tax” appears in the assistant-alignment literature as a warning that optimizing for helpfulness, harmlessness, and honesty can trade off against other capabilities (Askell et al., 2021); Ouyang et al. (2022) provide concrete InstructGPT evidence of benchmark regressions on tasks such as SQuAD, DROP, and HellaSwag. That evidence should not be read as a law of nature. It is an empirical frontier of a particular training recipe, model class, and evaluation set.

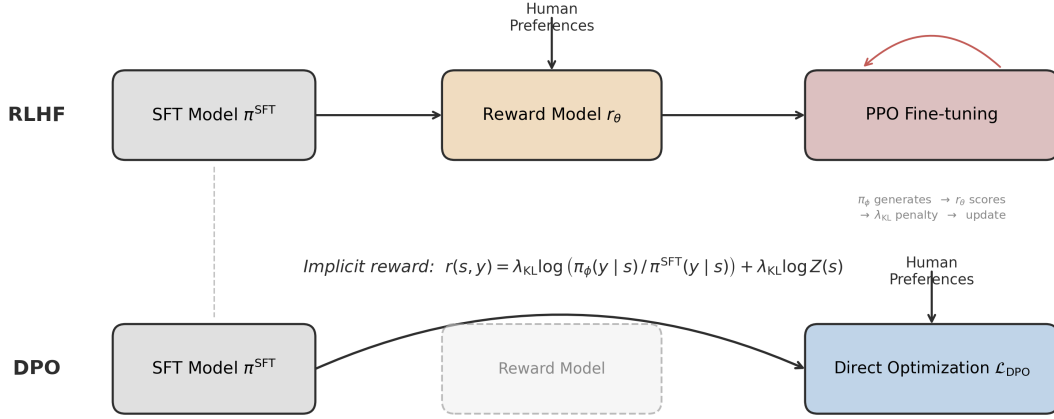


Figure 1: **RLHF versus DPO pipelines.** Top row: the three-stage RLHF pipeline trains a reward model from human preferences, then uses PPO to fine-tune the policy with a KL penalty. Bottom row: DPO collapses the pipeline into a single supervised learning objective over preference pairs, eliminating the explicit reward model (ghosted box).

The central welfare question is whose preferences are being aligned to. If one reward model is trained on pooled labels from a non-representative group of contractors, the resulting policy optimizes an implicit aggregation rule, not society’s preferences. Social choice supplies the vocabulary for this problem: alternatives, voters, ranking rules, impossibility theorems, strategic reports, minority protection, and interpersonal aggregation (Conitzer et al., 2024; Mishra, 2023). Distributional preference learning and axiomatic analyses show that standard BTL aggregation can encode Borda-like rules or fail desirable fairness and consistency conditions (Siththaranjan et al., 2024; Ge et al., 2024). Alternatives such as personalization, MaxMin-RLHF, Nash learning from human feedback, and maximal-lottery alignment make the aggregation rule more explicit, though some of these connections remain preprint-level rather than settled practice (Park et al., 2024; Chakraborty et al., 2024; Munos et al., 2023; Maura-Rivero et al., 2025).

0.4 Mechanism Design for Feedback

Feedback is not passively observed; it is elicited through a mechanism. Labelers may be inattentive, strategic, ideologically motivated, or selected by the platform’s contracting process. Mechanism design therefore enters before the reward model is trained. VCG-

style fine-tuning formulations ask how to aggregate multiple reward models when participants can benefit from manipulating the training rule (Sun et al., 2024). Strategyproof RLHF studies the same issue for preference reports, with a tradeoff between incentive compatibility and the quality of the final policy (Kleine Buening et al., 2025). Older peer-prediction mechanisms such as Bayesian truth serum and output-agreement methods show how payments can elicit truthful subjective signals when no ground truth is immediately observable (Prelec, 2004; Miller et al., 2005). For RLHF, this means the data-generating process is not a nuisance detail; it determines which preferences are observed and how costly truthful feedback is.

0.5 Structural Identification

Finally, preference optimization is not welfare analysis unless the learned reward has a structural interpretation. Bradley-Terry and DPO identify reward differences only through observed pairwise comparisons, up to normalizations and only on the covered comparison support. IRL identification results show that many rewards can rationalize the same behavior or preferences, and that entropy regularization, extra environments, or downstream invariance restrictions are needed to recover sharper objects (Cao et al., 2021; Skalse et al., 2023; Knox et al., 2022). Economically, the genealogy runs from McFadden logit choice (McFadden, 1974), through Rust-style dynamic discrete choice (Rust, 1994, 1996), to maximum-entropy IRL (Ziebart et al., 2008), and finally to DPO’s log-odds reparameterization (Rafailov et al., 2023). Recent econometric work makes this bridge explicit, but still as an emerging working-paper literature rather than a settled textbook synthesis (van der Laan et al., 2025). The practical lesson is conservative: RLHF can produce useful policies, but the reward object is partially identified, aggregation-dependent, and limited by the offline support of the comparison data.

0.6 Constitutional AI, RLAIF, and Scalable Oversight

Three departures from the canonical RLHF pipeline change the source of preference labels rather than the optimization objective. Constitutional AI (Bai et al., 2022) substitutes a short list of written principles for human harmlessness annotations, training the model to critique and revise its own outputs against the constitution and then distilling the resulting AI-generated comparisons into a preference model. RLAIF (Lee et al.,

2023) pushes the substitution one step further, using an off-the-shelf language model to label preference pairs directly, with reported parity against human-labelled RLHF on summarization and dialogue tasks and a direct variant that scores responses online during reinforcement learning. Scalable oversight (Bowman et al., 2022) operates one level up, asking whether human-model teams can supervise problems on which unaided humans underperform, with sandwiching experiments that bracket model capability between layperson and expert performance. Each method trades one cost for another, replacing large human-labelling budgets with the developer’s specification of principles, the labeler-model’s training distribution, or the design of human-AI interaction protocols. The substitution scrambles the identification question raised in the preceding subsections, since the encoded preferences no longer trace back to a labelled human-population sample but to whichever artifact is doing the labelling. Methodologically this sits in the same family as mechanism design with information acquisition, where the choice of signal technology changes which preferences are even observable.

0.7 Revealed Preference Tests and Axiom-Aware Aggregation

0.7.1 Revealed-preference tests of LLMs

A related literature treats language models themselves as economic agents and asks whether their choices satisfy revealed-preference restrictions. Budget-allocation experiments, consumer-preference replications, moral-choice tasks, and Bayesian decision problems find that LLMs can look rational in some narrow domains and fragile in others (Chen et al., 2023; Goli and Singh, 2024; Seror, 2024; Mu et al., 2025). These tests are useful because they translate vague alignment claims into falsifiable restrictions, such as GARP consistency or proximity to a Bayes rule. But the results are prompt-, model-, version-, and domain-sensitive. They should be used as diagnostics of a trained system, not as evidence that an LLM has a stable utility function in the welfare-economics sense.

The revealed-preference axioms invoked in these tests carry a long economics lineage. Samuelson (1938) introduced the weak axiom for choice over budget sets, Houthakker (1950) strengthened it to the strong axiom by closing transitivity over chains, Afriat (1967) proved that finite choice data admit a concave utility rationalization if and only if they satisfy GARP, and Varian (1982) laid out the nonparametric testing procedures that empirical revealed-preference work still uses. Transporting this machinery to language

models raises a translation question, since the classical theory presumes a single agent choosing across explicit budget sets while LLM tests treat prompt context as a budget analogue and generated token sequences as the chosen bundle. Standard RLHF further imposes Bradley-Terry pairwise consistency on the reward model, an assumption strictly stronger than GARP because it forces stochastic transitivity and a parametric latent reward (Raheja and Pochhi, 2026). A recent wave of methods relaxes this assumption along different axes. Liu et al. (2024) and Zhang et al. (2025) cast alignment as a two-player constant-sum game over policies and converge to a Nash policy that drops transitivity, accommodating cyclic and population-aggregated preferences. Tang et al. (2024) unifies DPO, IPO, and SLiC inside a convex family of offline losses and shows that the effective regularization differs from a global KL anchor, which reframes the question of which preference axioms each loss actually encodes. Gözl et al. (2025) imports Procaccia-style distortion bounds from social choice and proves that fitting a single Bradley-Terry reward model to heterogeneous user preferences incurs unavoidable welfare loss, while Nash Learning attains the minimax-optimal distortion bound. The empirical gap is that no current study runs GARP or SARP diagnostics directly on policies produced by these relaxed objectives, so the revealed-preference question for post-alignment language models is reopened rather than closed.

0.7.2 Linear social choice and the failure of Bradley-Terry MLE

Ge et al. (2024) cast the reward-modeling stage of RLHF as a social-choice problem in which a small set of candidates carry feature vectors and reward functions are linear in those features. Let $C = \{c_1, \dots, c_m\}$ be the candidate set, $x_c \in \mathbb{R}^d$ the feature vector of candidate c , and a parameter $\theta \in \mathbb{R}^d$ define the reward $r_\theta(c) = \langle \theta, x_c \rangle$. Each voter i has a private parameter θ_i whose induced ranking serves as their preference report; the aggregator observes pairwise comparisons sampled from these rankings and outputs a single θ^* . Two axioms from voting theory carry over.

Definition D1 (Pareto optimality). *A linear rank aggregation rule satisfies Pareto optimality (PO) if, whenever every voter ranks a above b in the input profile, the output ranking also places a above b .*

Definition D2 (Pairwise majority consistency). *A ranking σ is the pairwise majority consistent ranking for a profile if for every pair (a, b) , σ ranks a above b if and only if*

a majority of voters does. A rule satisfies PMC if it outputs this ranking whenever one exists and is feasible.

Theorem 1 (Theorem 3.1 of Ge et al. (2024)). *Any loss-based linear rank aggregation rule whose loss function is weakly convex and nondecreasing, or strictly convex, with $\inf_x \ell(x) < \ell(0)$, fails both Pareto optimality and pairwise majority consistency. The standard Bradley-Terry maximum-likelihood reward model is one such rule.*

The construction in the proof uses six candidates with feature vectors $x_a = (2, 1)$, $x_b = (1, 1)$, $x_c = (0, 0)$, $x_{a'} = x_a + (-\varepsilon, 0)$, $x_{b'} = x_b + (-\varepsilon, 0)$, $x_{c'} = x_c + (-\varepsilon, \delta\varepsilon)$, and two voter types whose linear parameters generate conflicting rankings. The PO violation appears at the pair (c', c) , on which both voter types agree but the BT-MLE optimum places c above c' for sufficiently small ε .

0.7.3 Leximax Copeland subject to Pareto optimality

To repair the axiomatic failure, Ge et al. (2024) propose a rule that adapts the classical Copeland voting rule to the linear-feature setting. For a profile π and pair (a, b) , let $w_{a \succ b}(\pi)$ denote the fraction of voters who rank a above b . The Copeland score of a is

$$C(a) = |\{b \in \mathcal{C} : w_{a \succ b}(\pi) > 1/2\}|.$$

The leximax Copeland rule ranks candidates by descending Copeland score, breaking ties by lexicographic comparison of the sorted margin vectors. Because not every ranking is feasible in the linear setting, the rule is implemented sequentially. For position $r + 1$ it picks the unranked candidate with the highest Copeland score for which there exists a parameter θ consistent with the partial ranking so far. The variant *leximax Copeland subject to PO* (LCPO) additionally enforces Pareto-dominance constraints, refusing to rank a Pareto-dominated candidate above its dominator. The whole rule reduces to a sequence of $\mathcal{O}(m^2)$ linear feasibility programs.

Theorem 2 (Theorem 4.3 of Ge et al. (2024)). *LCPO satisfies Pareto optimality, pairwise majority consistency, majority consistency, and winner monotonicity.*

0.7.4 Simulation Study: Axiom-Aware Aggregation on Heterogeneous Voters

We instantiate the construction above with $\varepsilon = 0.01$, $\delta = 2$, voter parameters $(1, 1)$ and $(-1, 0)$, population fractions $p = 0.6$ and $1 - p$, and Bradley-Terry label noise at scale $\beta = 50$. Two aggregators are compared. The BT-MLE aggregator fits a linear-in-features parameter θ by maximum likelihood on the pooled pairwise sample. The LCPO aggregator estimates pairwise majorities, computes Copeland scores, and applies the sequential feasibility procedure described above. We sweep $N \in \{5, 10, 20, 50, 100, 500, 2000\}$ comparisons per pair across thirty seeds, scoring outputs against the Pareto-dominance relation $\{(c', c)\}$ and the unique PMC ranking $a \succ a' \succ b \succ b' \succ c' \succ c$.

Figure 2 and Table 1 report the results. As N grows the BT-MLE aggregator violates both PO and PMC on every seed by $N = 2000$, exactly the asymptotic behavior predicted by Theorem 1, while LCPO satisfies PO on every seed at every sample size and satisfies PMC on every seed by $N = 100$. The worst-group utility of the top-ranked candidate converges to -2 for both methods. The axiomatic differential is therefore not an efficiency gain in the dominant winner but a guarantee on consistency with the elementary social-choice criteria that BT-MLE silently abandons under heterogeneous preferences.

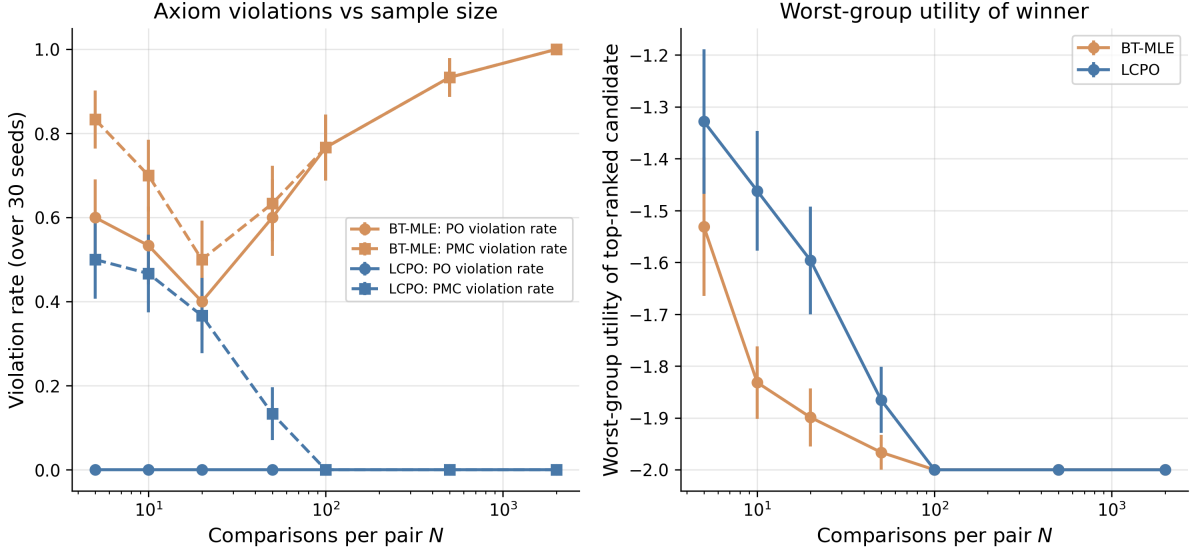


Figure 2: Axiom violation rates (left) and worst-group utility of the top-ranked candidate (right) versus pairwise comparisons per pair N , mean and one standard error over thirty seeds. BT-MLE PO and PMC violation rates rise to one as N grows, in line with Theorem 1; LCPO violation rates remain at zero (PO by construction) or fall to zero (PMC by $N = 100$).

Table 1: Aggregator comparison at $N = 2000$ comparisons per pair, mean and standard error over thirty seeds. PO viol. and PMC viol. are fractions of seeds on which the output ranking violates the corresponding axiom. Worst-group util. is the type-two utility of the top-ranked candidate. Modal top names the candidate most frequently ranked first.

Method	PO viol.	PMC viol.	Worst-group util.	Modal top
BT-MLE (linear)	1.00 (0.00)	1.00 (0.00)	-2.000 (0.000)	a
Leximax Copeland (LCPO)	0.00 (0.00)	0.00 (0.00)	-2.000 (0.000)	a

0.8 Simulation Study: Preference Learning in Job Search

A worker searches for jobs in a labor market with compensating differentials, following a McCall (1970)-style search model. Wages are measured in thousands and take values $w \in \{20, 28, 38, 50, 65, 82, 100, 125\}$, while amenities take values $z \in \{0, 1, \dots, 6\}$ and capture commute quality, flexibility, and job security. The state space has 112 states: 56 searching states in which the worker observes a pending offer (w_i, z_j) and decides to accept or reject, plus 56 employed states (w_i, z_j) in which the worker decides to stay or quit. The offer distribution exhibits compensating differentials, with wage rank and amenity rank negatively correlated ($\rho = -0.74$): high-wage offers cluster with low amenities and vice versa. The worker’s true per-period utility is $u(w, z) = \alpha \log(w) + (1 - \alpha)z$ with $\alpha = 0.6$, but this function is unobserved; the worker can only compare career trajectories (“I prefer path A to path B”), exactly as in stated-preference surveys in labor economics. While searching, the worker receives the unemployment benefit $u_b = \alpha \log(b)$ where $b = 28$. Layoffs occur with probability $p = 0.05$ per period, and the discount factor is $\gamma = 0.95$. Dynamic programming gives $V^*(s_0) = 74.13$; the optimal policy accepts 25 of 56 offer types and stays employed at 25 of 56 job types. Preference data is generated by rolling out a uniform random policy from a random searching state. Each rollout produces a career segment of $L = 15$ periods recording states and actions. For each of K comparisons, two independent career segments are generated; the segment with higher cumulative discounted utility under the true (unobserved) utility function is labeled as preferred via the Bradley-Terry model. Figure 3 shows the optimal accept/reject and stay/quit boundaries.

Six methods are compared across $K \in \{25, 50, 100, 200, 500, 1,000, 2,000, 5,000\}$, averaged over 30 seeds. The neural network RLHF reward model is a two-layer MLP trained by Bradley-Terry MLE on the K segment pairs; per-transition rewards are discount-

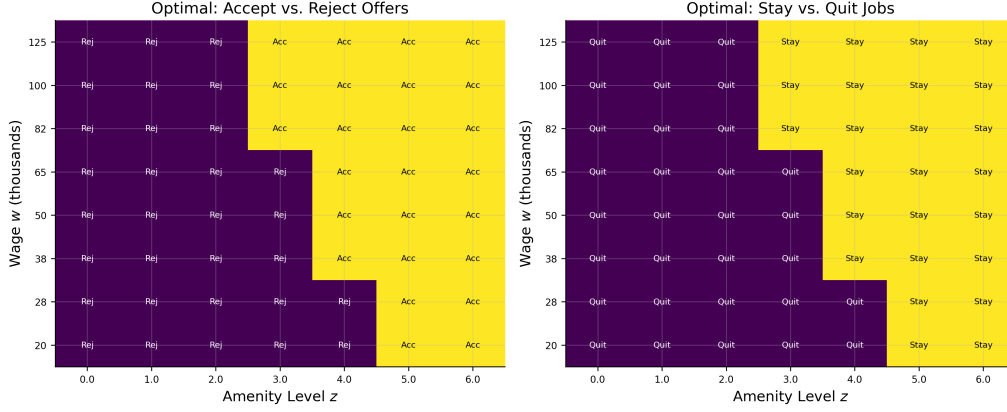


Figure 3: **Optimal accept/reject and stay/quit boundaries in wage-amenity space.** Left: the optimal accept/reject boundary for searching states. Right: the optimal stay/quit boundary for employed states. Both boundaries cut diagonally, reflecting the tradeoff between wage and amenity quality under compensating differentials.

weighted and summed to obtain a segment score, and the resulting 112-state reward table is solved by value iteration.⁶ The correctly specified structural model parameterizes utility as $\hat{u} = \hat{\alpha} \log(w) + (1 - \hat{\alpha})z$, estimating the single parameter $\hat{\alpha}$ via Bradley-Terry MLE, then solves the MDP. The misspecified model uses $\hat{u} = \hat{\alpha} \log(w) + (1 - \hat{\alpha})\bar{z}$, where $\bar{z} = 3$ is the mean amenity level; this model ignores amenity variation across jobs, treating all amenities as identical. DPO trains a tabular softmax policy directly from trajectory comparisons, bypassing reward modeling entirely.⁷ Tabular Q-learning (10,000 episodes, $\varepsilon = 0.15$, learning rate 0.1) and exact DP provide scalar-reward baselines. All four preference methods receive identical comparison data per seed; DPO receives a same-state variant generated from the same seed.

Figure 4 reports the main results. Three findings stand out. First, the correctly specified structural model dominates: even at $K = 25$ it achieves 99.9% of DP-optimal, illustrating the power of correct specification when the model has a single free parameter. Second, the neural network converges more slowly but reaches 99.9% by $K = 5,000$, reflecting the higher sample complexity of a flexible model relative to a one-parameter

⁶The neural network has 4 inputs (normalized log-wage, normalized amenity, employment indicator, action), 32 hidden units per layer, and $\sim 1,200$ parameters. The logistic loss from Equation (1) is applied to discount-weighted segment reward sums. The network parameterises $r_\theta(s, a)$, but the table handed to value iteration averages over actions to give $\bar{r}(s) = \frac{1}{2}(r_\theta(s, 0) + r_\theta(s, 1))$; this is harmless here because the true per-period reward in this environment is action-independent (actions affect only the next-state transition), so the Bradley-Terry MLE has no incentive to fit action-dependent payoffs.

⁷DPO uses 112 logit parameters ϕ_s , one per state, trained via the DPO loss (Equation 5) with Adam optimization over a sweep of $\lambda_{KL} \in \{0.01, 0.05, 0.1, 0.5, 1.0, 5.0\}$, selecting the λ_{KL} that minimizes training loss. The reference policy is uniform: $\pi^{SFT}(a|s) = 0.5$. To match the LLM setup where both completions condition on the same prompt, DPO comparison pairs start from the same initial state.

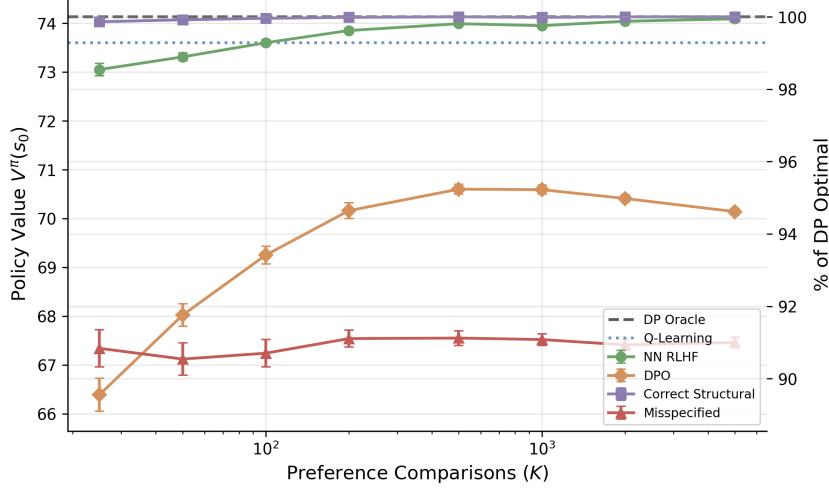


Figure 4: **Policy value $V^\pi(s_0)$ versus number of preference comparisons K for all six methods (30 seeds, $L = 15$).** The right axis shows the percentage of DP-optimal value.

structural specification. Third, DPO plateaus at approximately 95% by $K = 500$ and does not improve further.⁸ This plateau is qualitatively different from the gridworld failure (-118%): DPO recovers a reasonable policy, but the gap does not close with additional data.⁹ The misspecified constant-amenity model plateaus at 91% regardless of K , because additional preference data cannot correct the omitted variable.

Table 2: **Diagnostics at $K = 5,000$ (single seed): policy agreement with π^* , value-function correlation, and mean accepted wage and amenity for each method.**

Method	Policy agree. (%)	V^π corr.	Mean amenity	Mean wage	$\hat{\alpha}$
NN RLHF	96.4	1.000	4.67	73	—
DPO	57.1	0.777	3.38	63	—
Correct	100.0	1.000	4.84	71	0.597
Misspecified	50.0	0.889	3.00	70	—
Optimal	100.0	1.000	4.84	71	—

Table 2 provides state-level diagnostics at $K = 5,000$. The structural model and neural network achieve near-perfect policy agreement with π^* (100% and 96% respectively). DPO agrees on only 57% of states with value-function correlation 0.78, systematically underselecting on both wage and amenity dimensions.¹⁰ The misspecified model agrees

⁸DPO learns only from (s, a) pairs visited in training trajectories generated by the random behavioral policy. It cannot propagate value to unvisited states the way value iteration does after learning a reward model, so states poorly covered by the behavioral policy remain suboptimal regardless of K .

⁹DPO fails catastrophically in gridworld because transitions are stochastic (10% slip probability) and rewards are transition-dependent. The same (s, a) pair yields different rewards depending on whether the agent slipped, so the DPO loss conflates policy quality with transition luck. In the job search model, accept/reject deterministically changes employment status, and only the 5% layoff probability introduces stochastic transitions.

¹⁰DPO’s mean accepted amenity of 3.4 parallels the misspecified structural model’s 3.0, though the mechanisms

on 50%, with disagreements concentrated where amenity variation matters.¹¹

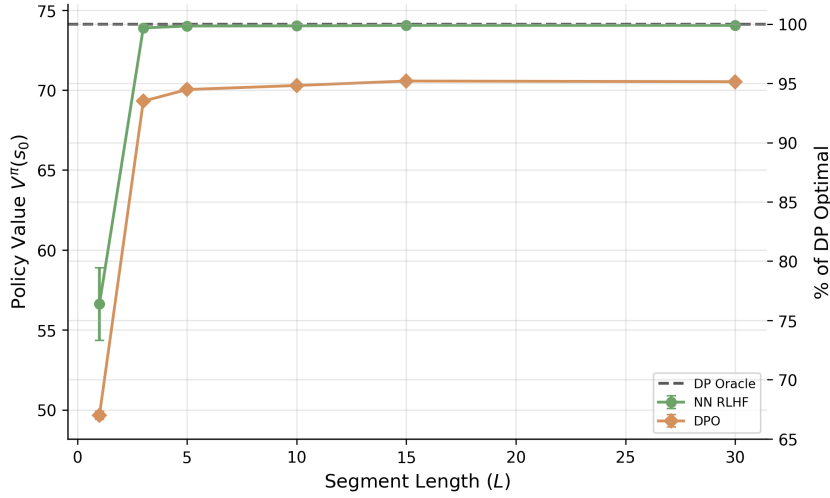


Figure 5: **Policy value $V^\pi(s_0)$ versus segment length L at $K = 2,000$ for the neural network and DPO (20 seeds).** The right axis shows the percentage of DP-optimal value.

Figure 5 reports the segment length ablation at $K = 2,000$. At $L = 1$, both methods perform poorly because single-step comparisons carry minimal information about long-run value. The neural network recovers rapidly (by $L = 3$) because the reward model aggregates per-transition estimates over longer segments and value iteration propagates them through the full transition structure; DPO improves monotonically but plateaus at its $\sim 95\%$ ceiling, as it must learn the policy directly from trajectory comparisons without access to the transition model.

Two-stage RLHF separates preference estimation (a static econometric problem) from dynamic programming (which exploits the known transition model); DPO conflates the two and forfeits the transition structure that structural modelers typically have access to.¹² The DPO plateau in the job-search study is the computational version of the identification point above. More comparisons tighten the empirical binary choice model on the support of observed trajectories, but they do not reveal counterfactual value in poorly covered states, nor do they identify a welfare object independent of the aggregation

differ: the misspecified model ignores amenity variation by construction, while DPO underweights amenities because the random behavioral policy underrepresents high-amenity employed states in the training data.

¹¹An online versus offline ablation for the neural network at $K = 1,000$ (20 seeds) shows comparable performance: online 73.92 ± 0.05 , offline 73.99 ± 0.03 ($p = 0.09$). The random behavioral policy already provides diverse career trajectories covering the full wage-amenity space.

¹²This advantage is specific to settings where the transition model is known or estimable; in domains without a tractable transition model, DPO’s single-stage approach avoids compounding errors from reward model estimation.

rule used to construct the labels.

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